Urban Footprint Intelligence: Predicting Commercial Real Estate Vacancy Risk Using Public Mobility and Urban Infrastructure Data

Bogapriya Muralikannan (ID: NF1015525)

Master of Data Analytics

DAMO 699: Capstone Project Section 1

Professor Mehdi Mostofi

Capstone Project Proposal

1. Introduction

Commercial real estate markets have undergone massive structural shifts due to evolving hybrid work patterns, economic fluctuations, and shifting urban migration. For property managers, municipal economic teams, and small business owners, understanding which commercial zones or storefront locations face a high risk of prolonged vacancy is critical for local financial planning and economic revitalization.

Rather than relying solely on traditional lagging financial indices or retroactive lease termination alerts, this project introduces a proactive framework by leveraging open-source urban mobility indicators, public transit proximity, neighborhood demographic shifts, and local business density to evaluate and predict geographic property risk.

2. Problem Statement

Many municipal economic development teams and commercial property stakeholders lack a data-driven, localized framework to evaluate neighborhood business stability in real time. Traditional vacancy assessments are reactive, occurring only after a business closes or a storefront sits dark. This lack of predictive visibility makes it incredibly difficult to implement proactive localized economic interventions, optimize commercial zoning, or mitigate financial losses due to empty properties.

This project aims to address whether open-source urban infrastructure, spatial business density metrics, and neighborhood demographic data can be engineered to predict commercial vacancy risk zones before vacancies peak.

3. Research Questions

Primary Research Question

- Can localized urban infrastructure and surrounding business density features be utilized to accurately classify the vacancy risk level of commercial business zones?

Secondary Research Questions

- Which urban factors (e.g., proximity to public transit, local walkability indicators, or specific competitor density) have the strongest statistical relationship with commercial stability?
- Does an over-saturation of similar business types within a specific geographic radius significantly increase the risk of neighborhood vacancy?
- Which machine learning classification algorithm provides the most accurate and interpretable risk assignment?

4. Research Hypotheses

- Null Hypothesis (H_0): Public urban infrastructure metrics, pedestrian transit accessibility, and surrounding business density features have no statistically significant relationship with commercial vacancy rates.
- Alternative Hypothesis (H_1): Localized urban infrastructure and surrounding business density features have a statistically significant relationship with commercial vacancy rates and can accurately classify high-risk zones.

5. Project Objectives

- Acquire and Clean Open Data: Gather public business licensing records and urban infrastructure mapping data from Canadian municipal open data portals.

- Execute Spatial Feature Engineering: Calculate distance-to-transit metrics, local business diversity indices, and temporal growth rates of new business openings versus closures.
- Develop a Risk Scoring Model: Build a machine learning classification pipeline to segment municipal zones or business clusters into Low, Medium, and High vacancy risk levels.
- Deploy an Interactive Dashboard: Construct a functional Map-Based Streamlit dashboard showing a "Municipal Economic Health Index" for stakeholders to actively monitor geographic risk.

6. Data Specifications

The project will rely entirely on free, open-source civic data platforms, ensuring data accessibility and reproducibility:

- Municipal Business Licensing Datasets: Sourced via Canadian municipal portals (such as Toronto Open Data or Open Data Hamilton). These provide historical records of active, renewed, or canceled business licenses, which will serve as our proxy for commercial active/vacant status.
- Public Transit Infrastructure Data (GTFS): Spatial locations and geographical coordinates of transit stops, subway stations, and public parking infrastructure.
- Statistics Canada Neighborhood Profiles: Open census-level demographic data (such as localized population growth and median income) mapped to local forward sortation areas (FSAs) or neighborhood clusters.

7. Proposed Methodology

The project will follow a structured four-phase data analytics lifecycle:

Phase 1: Data Collection & Integration

- Extract municipal business registry records, census-level tables, and infrastructure spatial coordinates in CSV and GeoJSON formats.
- Clean missing data, standardize business categories (e.g., retail, food service, professional offices), and derive active versus inactive status to construct the target variable.

Phase 2: Data Cleaning & Feature Engineering

- Competition Density: Compute the volume of active businesses in the exact same industry sector within a localized 1 km radius using spatial coordinates.
- Transit Accessibility: Calculate the exact distance (in meters) from a business cluster to the nearest major public transit hub.
- Area Churn Rate: Calculate the ratio of canceled licenses to active licenses in a specific neighborhood over the past 24 months.

Phase 3: Exploratory Data Analysis & Modeling

- Perform exploratory data analysis to evaluate spatial distributions and feature correlations.
- Train classification models to predict whether an area's business stability will decline into a high-vacancy risk zone.
- Algorithms to Test: Logistic Regression, Random Forest, and Gradient Boosting (XGBoost).
- Evaluation Metrics: Precision, Recall, and F1-Score (prioritizing high recall to prevent missing high-risk zones).

Phase 4: Dashboard Development & Deployment

Develop an interactive Streamlit dashboard integrated with geospatial plotting libraries (folium or plotly). The dashboard will feature:

- An interactive map color-coded by Economic Risk Level (Green = Low, Yellow = Medium, Red = High Risk).
- Key metric widgets showing feature importance (e.g., demonstrating how "Distance to Transit" impacts local stability).
- A lookup tool where a user can select a neighborhood and instantly view its stability forecast.

8. Expected Deliverables

- A structured, engineered spatial dataset of municipal business zones.
- A reproducible Python data cleaning and machine learning pipeline.
- An exploratory data analysis report mapping urban indicators.
- An interactive, web-based risk analytics dashboard.
- A comprehensive final capstone report and project demonstration.

9. Tools and Technologies

- Language: Python (pandas, numpy, scikit-learn, folium / plotly)
- Database/Storage: SQL for data aggregation and cleaning
- Visualization/App Layer: Streamlit or Power BI
- Version Control: Git / GitHub